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Course Name: COMPUTER VISION

C02-AT1-Analytical Problem Solving

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

Grey level transformations operate directly on pixels to modify image contrast and brightness. They are essential for emphasizing specific ranges of grey levels, inverting images (like negative transformations) for easier medical visual analysis, and compressing dynamic ranges to reveal hidden details.

(b) Apply a negative transformation $s = 255 - r$ to the pixel values: $r = [50, 100, 150, 200]$. Compute the transformed values.

Calculation:

$$s_1 = 255 - 50 = 205$$

$$s_2 = 255 - 100 = 155$$

$$s_3 = 255 - 150 = 105$$

$$s_4 = 255 - 200 = 55$$

Output: The transformed values are [205, 155, 105, 55].

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

Log transformation expands the values of dark (low-intensity) pixels while simultaneously compressing the higher-level values. This selective expansion increases the contrast in darker regions, making hidden details in shadows or under-illuminated areas visible.

(b) Given $s = c * \log(1 + r)$, where $c = 10$, compute s for: $r = [0, 10, 50, 100]$

Calculation (Assuming base-10 logarithm):

$$r = 0 \Rightarrow s = 10 * \log_{10}(1) = 0$$

$$r = 10 \Rightarrow s = 10 * \log_{10}(11) = 10 * 1.0414 = 10.41$$

$$r = 50 \Rightarrow s = 10 * \log_{10}(51) = 10 * 1.7075 = 17.08$$

$$r = 100 \Rightarrow s = 10 * \log_{10}(101) = 10 * 2.0043 = 20.04$$

Output: The computed transformed values are roughly [0, 10.41, 17.08, 20.04].

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.

Histogram equalization spreads out the most frequent intensity values, flattening the overall histogram distribution. This effectively maps a narrow range of input intensities to a wider range, thereby maximizing the global contrast of the image.

(b) Given a 3-bit image with given histogram. Compute the cumulative distribution and equalized intensity values.

Calculation:

Frequencies: {0:2, 1:2, 2:4, 3:8}. Total pixels $N = 16$.

Equalized intensities formula: $s_k = \text{round}(7 * \text{CDF}(k) / N)$

$k=0$: $\text{CDF} = 2 \Rightarrow s_0 = \text{round}(7 * 2/16) = \text{round}(0.875) = 1$

$k=1$: $\text{CDF} = 2+2=4 \Rightarrow s_1 = \text{round}(7 * 4/16) = \text{round}(1.75) = 2$

$k=2$: $\text{CDF} = 4+4=8 \Rightarrow s_2 = \text{round}(7 * 8/16) = \text{round}(3.5) = 4$

$k=3$: $\text{CDF} = 8+8=16 \Rightarrow s_3 = \text{round}(7 * 16/16) = \text{round}(7.0) = 7$

Output: Equalized intensities for {0, 1, 2, 3} are {1, 2, 4, 7}.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Spatial smoothing filters act as low-pass filters that blur the image. They are primarily used to reduce noise, remove small irrelevant details, and bridge small gaps in lines or curves before structural analysis.

(b) Apply a 3x3 averaging filter to the following region: [10 20 30; 20 30 40; 30 40 50]. Compute the new center pixel value.

Calculation:

Sum of all 9 pixels = $10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$

Averaging mask weight = $1/9$

New Center Value = $270 / 9 = 30$

Output: The new center pixel value is 30.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Sharpening acts as a high-pass spatial filter. It highlights intensity transitions (edges) and fine details by using mathematical derivatives (such as the Laplacian) to increase the contrast between adjacent areas of different intensities.

(b) Apply the Laplacian mask to the matrix: [10 10 10; 10 20 10; 10 10 10]. Compute the output at the center.

Calculation:

Center overlap = $4 * 20 = 80$

Top, Bottom, Left, Right overlaps = $(-1 \times 10) + (-1 \times 10) + (-1 \times 10) + (-1 \times 10) = -40$

Corner overlaps = $(0 \times 10) = 0$

Sum = $80 - 40 + 0 = 40$

Output: The output at the center is 40.

6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

A low-pass filter preserves low-frequency components (smooth regions) while attenuating high frequencies, resulting in image blurring. A high-pass filter preserves high-frequency components (edges) while attenuating low frequencies, resulting in sharpened images.

(b) Given frequency components: $F = [100, 80, 40, 20]$. Apply a low-pass filter that retains only the first two components. Write the filtered output.

Calculation:

The filter acts as an ideal low-pass filter cutoff that keeps the first two elements and zeros out the rest.

Output: The filtered output is $[100, 80, 0, 0]$.

7. Thresholding

(a) Explain global thresholding in segmentation.

Global thresholding separates an image into foreground and background by comparing each pixel's intensity to a single, constant threshold value T across the entire image.

(b) Given pixel values: $[200]$. Apply threshold $T = 100$. Assign values (0 or 1).

Calculation:

Pixel intensity = 200.

Condition: Is $200 \geq 100$? Yes.

Output: The assigned value is 1.

8. Edge Detection

(a) Explain the concept of edge detection.

Edge detection identifies points in a digital image where luminous intensity changes sharply. It captures important structural properties by finding boundaries between distinct regions.

(b) Apply the Sobel operator (horizontal) to: $[10 \ 10 \ 10; 20 \ 20 \ 20; 30 \ 30 \ 30]$.

Compute the gradient at the center.

Calculation:

Top row overlap: $(-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -10 - 20 - 10 = -40$

Middle row overlap: $(0*20) + (0*20) + (0*20) = 0$

Bottom row overlap: $(1*30) + (2*30) + (1*30) = 30 + 60 + 30 = 120$

Gradient = $-40 + 0 + 120 = 80$

Output: The gradient at the center is 80.

9. Region-Based Segmentation

(a) Explain region growing in segmentation.

Region growing is a procedure that groups pixels into larger regions based on a pre-defined similarity criterion. It starts with a 'seed' point and expands by iteratively appending neighboring pixels that have properties similar to the seed within a defined threshold.

(b) Given pixel intensities: [1 0 2]. If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

Calculation:

Valid region range: $[\text{Seed} - 5, \text{Seed} + 5] = [45, 55]$

Evaluate pixel 1: 1 not in $[45, 55]$

Evaluate pixel 0: 0 not in $[45, 55]$

Evaluate pixel 2: 2 not in $[45, 55]$

Output: None of the pixels belong to the region.

10. Combined Enhancement & Segmentation

(a) Explain why smoothing is applied before segmentation.

Smoothing serves as a critical preprocessing step to reduce high-frequency noise.

Without it, noise could be incorrectly identified as edges or cause over-segmentation.

(b) Given image values: [1 0 5]. First apply averaging (assume mean = 48), then threshold at $T = 50$. Determine final segmented output.

Calculation:

Step 1: Smoothing output is provided as the mean = 48.

Step 2: Apply threshold $T = 50$.

Condition: Is $48 \geq 50$? No.

Output: The final segmented output is 0.